

Single Organ Segmentation (Model U-Net++)

(Intestinal Veins, Ureter, Inferior mesenteric artery (IMA), Vesicular glands (Vglands), Pancreas)

(Framework=PyTorch, Encoder=resnet50, Treshold=0.40, trainig size=0.8, test_size=0.1, validation size=0.1)

a) **Intestinal Veins Data** (Train image=**853**, Test image =**108**, validation image=**106**)

image size 128x128 (Model U-Net++)

Number of Epochs	10 Epoch	30 Epoch	50 Epoch
IoU	0.4683	0.5344	0.5564
Precision	0.5579	0.6541	0.7153
Recall	0.7474	0.7929	0.6948
Specificity	0.9942	0.9957	0.9973
F1 Skore	0.6078	0.6961	0.6829
Running Time (seconds)	346.62	1017.87	1700.47

image size 256x256 (Model U-Net++)

Number of Epochs	10 Epoch	30 Epoch	50 Epoch
IoU	0.5756	0.6256	0.6400
Precision	0.6398	0.7311	0.7810
Recall	0.8533	0.7948	0.7770
Specificity	0.9949	0.9968	0.9978
F1 Skore	0.7102	0.7497	0.7614
Running Time (seconds)	550.43	1562.75	2580.05

b) **Ureter Data** (Train image=**1020**, Test image=**128**, validation image=**127**)

image size 128x128 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.5135	0.5340	0.5574
Precision	0.6267	0.6615	0.7020
Recall	0.6917	0.7134	0.6849
Specificity	0.9966	0.9966	0.9978
F1 Skore	0.6293	0.6577	0.6710
Running Time (seconds)	389.98	1159.15	1865.24

image size 256x256 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.5479	0.5666	0.5976
Precision	0.6467	0.7400	0.7657
Recall	0.7687	0.6872	0.7361
Specificity	0.9954	0.9976	0.9980
F1 Skore	0.6736	0.6849	0.7144
Running Time (seconds)	619.81	1430.20	1980.77

c) **Inferior mesenteric artery Data** (Train image=872, Test image=109, validation image=109)

image size 128x128 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.4991	0.5461	0.5550
Precision	0.7003	0.6939	0.7486
Recall	0.6212	0.7055	0.6851
Specificity	0.9939	0.9932	0.9946
F1 Skore	0.6251	0.6763	0.6862
Running Time (seconds)	369.46	1036.08	2410.75

image size 256x256 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.5223	0.5933	0.6304
Precision	0.7080	0.6969	0.7650
Recall	0.6561	0.7884	0.7774
Specificity	0.9939	0.9918	0.9945
F1 Skore	0.6515	0.7222	0.7553
Running Time (seconds)	543.06	1310.96	2546.62

d) **Vglands Data** (Train image=958, Test image=121, validation image=119)

image size 128x128 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.4151	0.4581	0.5319
Precision	0.5404	0.6397	0.7375
Recall	0.5944	0.5961	0.6470
Specificity	0.9900	0.9940	0.9962
F1 Skore	0.5286	0.5747	0.6463
Running Time (seconds)	446.91	1199.17	2695.07

image size 256x256 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.4871	0.5066	0.5983
Precision	0.5755	0.7033	0.7783
Recall	0.7411	0.6439	0.7124
Specificity	0.9896	0.9941	0.9967
F1 Skore	0.6105	0.6264	0.7079
Running Time (seconds)	638.21	1872.66	2755.72

e) **Pancreas Data** (Train image=938, Test image=118, validation image=117)

image size 128x128 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.5384	0.5518	0.5736
Precision	0.6464	0.6884	0.7392
Recall	0.7468	0.7184	0.7004
Specificity	0.9925	0.9937	0.9960
F1 Skore	0.6628	0.6811	0.6934
Running Time (seconds)	376.13	1075.29	2677.30

image size 256x256 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.6069	0.6304	0.6324
Precision	0.7336	0.7730	0.7531
Recall	0.7639	0.7610	0.7786
Specificity	0.9944	0.9958	0.9950
F1 Skore	0.7281	0.7523	0.7532
Running Time (seconds)	607.93	1581.08	2758.55

image size 512x512 (Model U-Net++)

Number of Epochs	10	30	50
IoU	0.5498	0.6546	0.6675
Precision	0.5993	0.8048	0.8235
Recall	0.8208	0.7779	0.7654
Specificity	0.9882	0.9958	0.9964
F1 Skore	0.6745	0.7601	0.7765
Running Time (seconds)	732.72	3076.99	6451.02